Reg. No. | | | | | | | | | | | [17MT42]

FOURTH SEMESTER B.TECH EXAMINATION, JUNE-2019
MINERALS PROCESSING

Time: 3 Hours **Maximum Marks:70**

Instructions:
1. Answer any **FIVE** full Questions by choosing at least **TWO** questions from each part.
2. Missing data may be suitably assumed.
3. Use of graph sheets is permitted.

PART-A

1. a. Discuss briefly the scheme of mineral processing. **07**

b. Differentiate between ore and mineral. Mention the ore names and its chemical formula for the following metals **07**
- **i.** Fe
- **ii.** Pb
- **iii.** Cu
- **iv.** Al
- **v.** Ni.

2. a. Define shape factor. Show that shape factor of a cuboidal particle is 1.24. **06**

b. Explain the operation of a gyratory crusher using a neat diagram. Mention its merits over jaw crusher. **08**

3. a. Describe the working principle of a ball mill. **07**

b. Illustrate the determination of 80% passing size (d_{80}) from the sieve analysis. **07**

4. a. What is settling ratio? Derive free-settling and hindered settling ratios for a mixture of fine quartz (density 2.65 g/cc) and galena (density 7.5 g/cc) particles. Assume a pulp density of 1.5 g/cc for hindered settling. **06**

b. Derive the mathematical expression for ratio of concentration, ratio of recovery and ratio of enrichment. Show the relationship between them. **08**

PART-B

5. a. Explain the principle of jigging. **06**

b. Describe the following with neat sketches **08**
- **i.** Reichert cones
- **ii.** Flowing film concentration

6. a. Explain the operation of induced roll magnetic separator with a neat sketch. **07**

b. Discuss briefly about high tension separator. **07**

7. a. Define the following: **06**
- **i.** Direct flotation
- **ii.** Reverse flotation
- **iii.** Differential flotation
- **iv.** Bulk flotation

b. Illustrate column flotation with flow sheet. **08**

8. a. A double roll crusher with rolls of 140 Cm diameter and 50 Cm width can crush the particles to a size of 60 Cm. Calculate the maximum product size if the co-efficient of friction between roll and particle surface is 0.28. Compare the results when the co-efficient of friction is 0.32. **06**

b. Draw a generalised flow sheet for mineral beneficiation. **08**

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